MATHS MANTHAN
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**Subject:** Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 07** **Time:** 60 Minutes
**Class:** Class X (CBSE) **Chapter 10: Circles** **MM: 30**

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**GENERAL INSTRUCTIONS:**

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

## SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

**Q1.** From a point $P$, two tangents $PA$ and $PB$ are drawn to a circle with centre $O$. If $\angle APB = 50°$, then $\angle AOB$ is: [1]

(A) $100°$ (B) $130°$ (C) $140°$ (D) $150°$

**Q2.** In a right triangle $ABC$ right-angled at $C$, $BC = 5$ cm and $AC = 12$ cm. The inradius $r$ of the incircle is: [1]

(A) 1.5 cm (B) 2 cm (C) 2.5 cm (D) 3 cm

**Q3.** If chord $AB$ of length 16 cm of a circle of radius 10 cm is tangent to an inner concentric circle, then the radius of the inner circle is: [1]

(A) 5 cm (B) 6 cm (C) 7 cm (D) 8 cm

**Q4.** If tangents $PA$ and $PB$ are drawn to a circle with centre $O$ such that $\angle APB = 120°$, then the relation between $OP$ and tangent $AP$ is: [1]

(A) $OP = AP$ (B) $OP = 2AP$ (C) $OP = \sqrt{3}AP$ (D) $AP = 2OP$

**Q5.** A line that intersects a circle in two distinct points is called a: [1]

(A) Tangent (B) Secant (C) Radius (D) Normal

**Q6.** **Assertion (A):** If two tangents $TP$ and $TQ$ are drawn from an external point $T$ to a circle with centre $O$, then $\angle PTQ = 2\angle OPQ$.
**Reason (R):** In $\triangle TPQ$, $TP = TQ$ implies $\angle TPQ = \angle TQP = \frac{1}{2}(180° - \angle PTQ)$, and $\angle OPT = 90°$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

## SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

**Q7.** A chord of length 16 cm of a circle of radius 10 cm is tangent to an inner concentric circle. Find the radius of the inner circle. [2]

**Q8.** If $PA$ and $PB$ are two tangents drawn from an external point $P$ to a circle with centre $O$ such that $\angle APB = 60°$ and $PA = 7$ cm, find the length of chord $AB$. [2]

## SECTION C: Short Answer Type (2 × 3 = 6 Marks)

**Q9.** Two concentric circles are of radii $R$ and $r$ $(R > r)$. A chord $AB$ of the outer circle touches the inner circle at $P$. Prove that the length of chord $AB$ is $2\sqrt{R^2 - r^2}$. If $R = 13$ cm and chord length is 24 cm, find $r$. [3]

**Q10.** In an isosceles $\triangle ABC$ with $AB = AC = 10$ cm and $BC = 12$ cm, a circle is inscribed. Find the lengths of the tangent segments into which the sides are divided by the points of contact. [3]

## SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

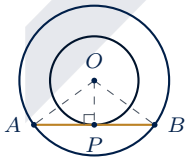
**Q11.** A triangle $ABC$ is drawn to circumscribe a circle of radius 4 cm such that the segments $BD$ and $DC$ into which $BC$ is divided by the point of contact $D$ are of lengths 8 cm and 6 cm respectively. Find the sides $AB$ and $AC$. [5]

**Q12.** Prove that the opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle. Using this result, if $\angle AOB = 125°$ where $O$ is the centre, find the measure of $\angle COD$. [5]

## SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

**Q13. Concentric Circular Running Track and Direct Sprint Chords**

A sports stadium features two concentric circular tracks of inner radius $r = 20$ m and outer radius $R = 25$ m sharing a common centre $O$. A straight sprint corridor $AB$ is painted along a chord of the outer track such that it just grazes (is tangent to) the inner track at point $P$.




(i) State the angle between radius $OP$ and the sprint chord $AB$. [1]

(ii) Find the length $AP$ of the half-chord. [1]

(iii) Calculate the total length of the sprint corridor $AB$, OR find the area of the track region between the concentric boundaries. [2]

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